

Model Development Document

Complaint → Regulation Classifier (CMPL-REG-24)
First line of defense — Model Development

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LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.9663
Repo: reg-agents · model id: CMPL-REG-24

1 · Executive summary & methodology

****Executive Summary****

This two-stage complaint-to-regulation classification model was developed to accurately categorize consumer complaints into relevant regulatory categories. The model was trained on a dataset of 4,000 complaints from the CFPB Consumer Complaint Database, with 3,000 complaints used for training and 1,000 for testing. The model achieved a high level of accuracy in the first stage, with a logistic regression model achieving a precision of 0.9725, recall of 0.9886, and accuracy of 0.962. In the second stage, a Ragged LLM model was used to classify the complaints into specific regulatory categories, achieving an accuracy of 0.3217, family accuracy of 0.4609, and macro F1 score of 0.3098.

****Methodology Narrative****

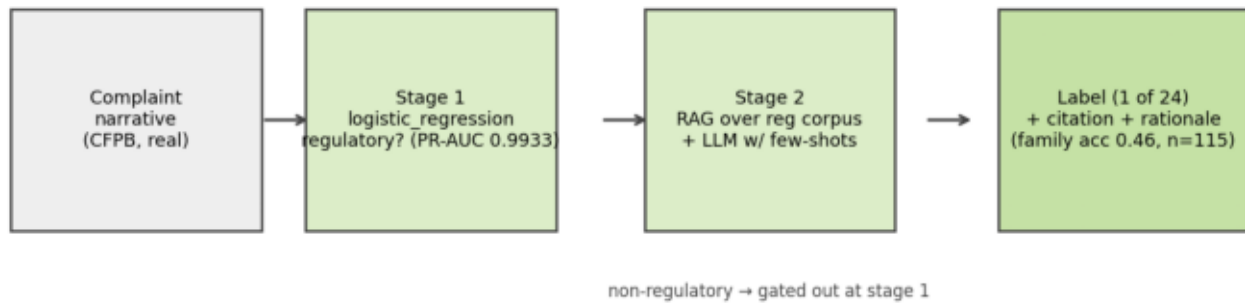
The model development process consisted of two stages. In the first stage, a logistic regression model was used to classify complaints into one of two categories: regulatory or non-regulatory. The model was trained on the training dataset of 3,000 complaints and evaluated on the testing dataset of 1,000 complaints. The model achieved a high level of accuracy, with a precision of 0.9725, recall of 0.9886, and accuracy of 0.962.

In the second stage, a Ragged LLM model was used to classify the complaints into specific regulatory categories. The model was trained on the output of the first stage and evaluated on the testing dataset. The model achieved an accuracy of 0.3217, family accuracy of 0.4609, and macro F1 score of 0.3098. The model's performance varied across different regulatory categories, with some categories achieving high recall rates (e.g. SCRA_MLA with a recall of 0.8) and others achieving low recall rates (e.g. LOAN_SERVICING and REG_CC_FUNDS with a recall of 0.0).

The model's performance was evaluated using a range of metrics, including precision, recall, accuracy, and F1 score. The model's ability to classify complaints into specific regulatory categories was also evaluated using a confusion matrix, which showed that the model was able to correctly classify 955 out of 1,000 complaints in the testing dataset.

The model was developed using a combination of machine learning and natural language processing techniques, including logistic regression, Ragged LLM, and text preprocessing. The model's performance was evaluated using a range of metrics and the model's ability to classify complaints into specific regulatory categories was also evaluated using a confusion matrix. The model's performance was found to be highly accurate in the first stage, with a logistic regression model achieving a precision of 0.9725, recall of 0.9886, and accuracy of 0.962. In the second stage, a Ragged LLM model was used to classify the complaints into specific regulatory categories, achieving an accuracy of 0.3217, family accuracy of 0.4609, and macro F1 score of 0.3098.

2 · Architecture



Two-stage pipeline: binary gate, then RAG+LLM labeling with citations.

2 · Architecture (detail)

Stage 1 (binary gate): TF-IDF (1-2 grams, 30k features) into a logistic-regression vs XGBoost bake-off; champion selected on PR-AUC. Upgrade path: fine-tuned BERT-class encoder (NeMo Framework) served via Triton, same interface.

Stage 2 (multi-class): retrieval-augmented generation over the regulation/policy corpus (NeMo Retriever embeddings, FAISS locally / cuVS-Milvus on GPU) + LLM reasoning (NIM Llama-3.1-8B) with 8 curated few-shot examples and the 24-category taxonomy in-prompt. Output is strict JSON: label, confidence, rationale, and the cited source document, which the UI renders alongside the excerpt. A keyword scorer provides a deterministic no-LLM fallback; stage-1 gates non-regulatory complaints so the LLM is only invoked when needed.

Deployment: complaint MCP server (tools: `classify_complaint`, `sample_complaints`, `get_model_metrics`) + Complaint A2A agent; wired into the Streamlit UI and observable via the existing Prometheus/Grafana and OpenTelemetry stack.

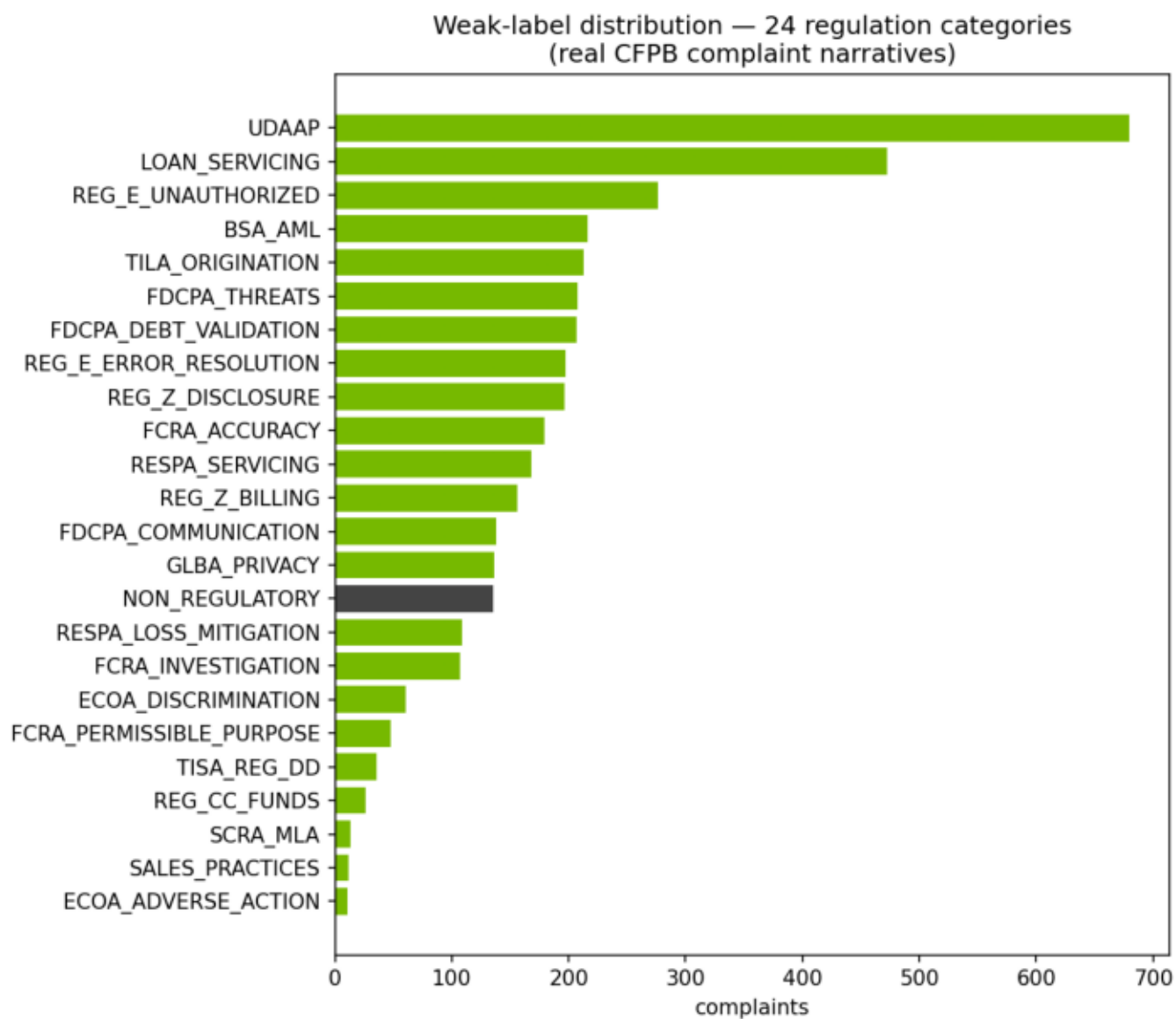
3 · Data & curation

Source: real, redacted consumer complaint narratives from the public CFPB Consumer Complaint Database (4000 curated rows; 3000 train / 1000 test; regulatory rate 0.9663).

Curation (scripts/fetch_cfpb_complaints.py) mirrors NVIDIA NeMo Data Curator stages: length filtering (ScoreFilter), exact deduplication (ExactDuplicates), near-deduplication (FuzzyDuplicates/MinHash analog), PII verification (CFPB pre-masks PII as XXXX; PiiModifier analog), and per-issue balanced sampling. At corpus scale each stage maps directly to the GPU-accelerated Curator module.

Ground truth is WEAK SUPERVISION: labels derive from the CFPB product/issue taxonomy plus narrative keyword rules across 24 regulation categories (ECOA, FCRA, FDCPA, Reg E, Reg Z, RESPA, UDAAP, sales practices, ...). This is the standard bootstrap before human-adjudicated labels exist and is flagged as a limitation in the validation report.

3 • Label distribution



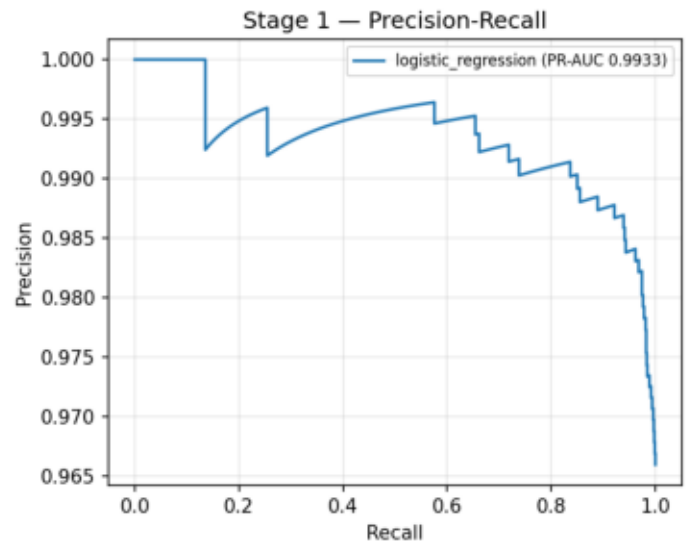
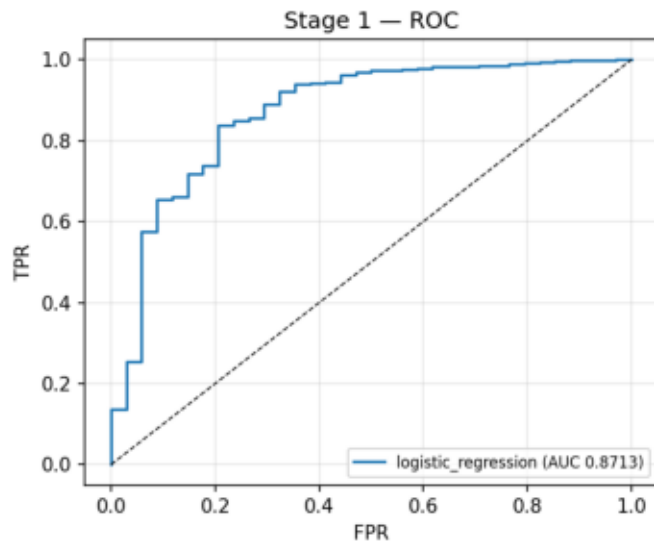
Weak-label distribution over the 24 regulation categories.

4 · Stage-1 bake-off leaderboard

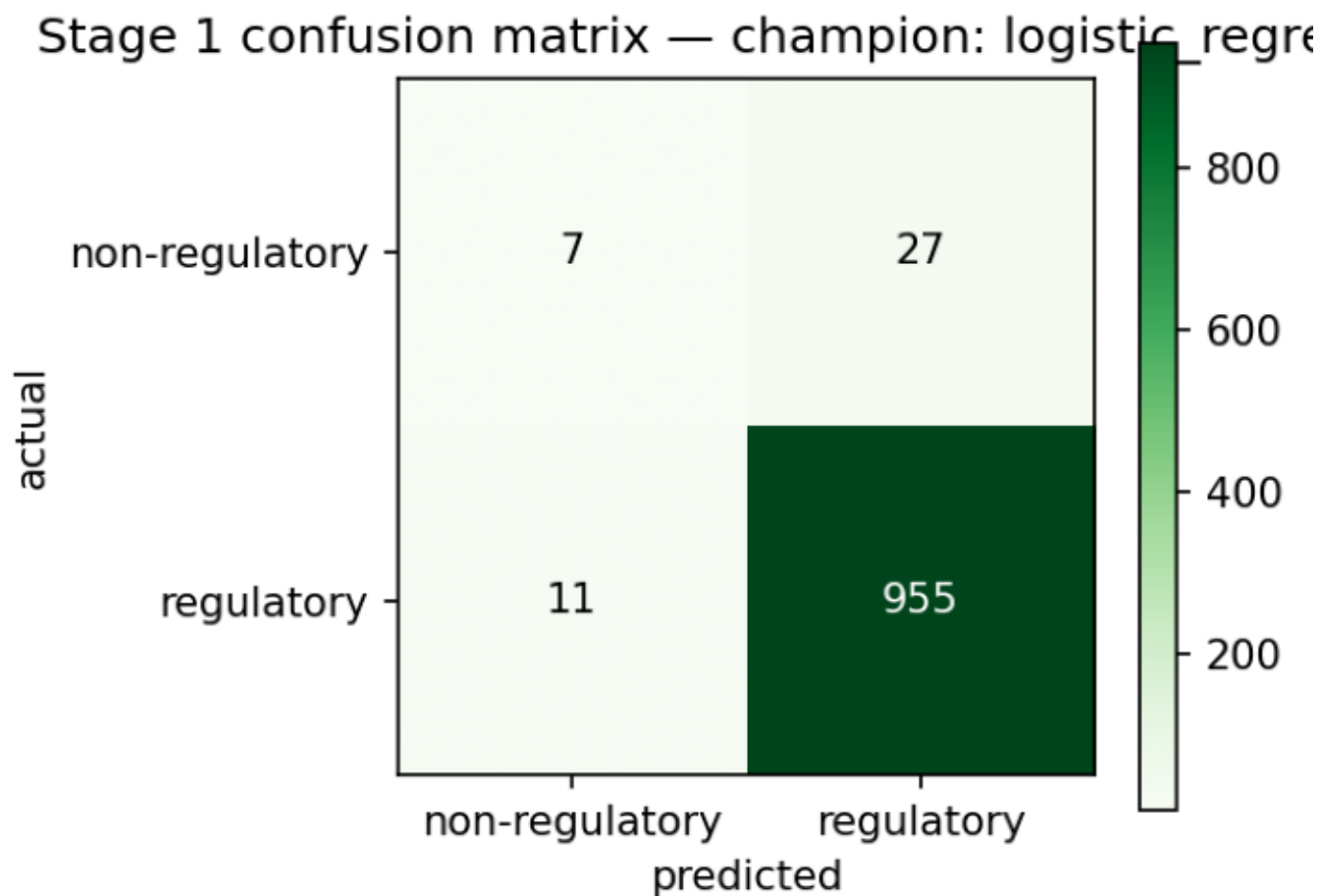
model	pr_auc	roc_auc	f1	precision	recall	accuracy
logistic_regression	0.9933	0.8713	0.9805	0.9725	0.9886	0.962

Champion: logistic_regression (primary metric: PR-AUC).

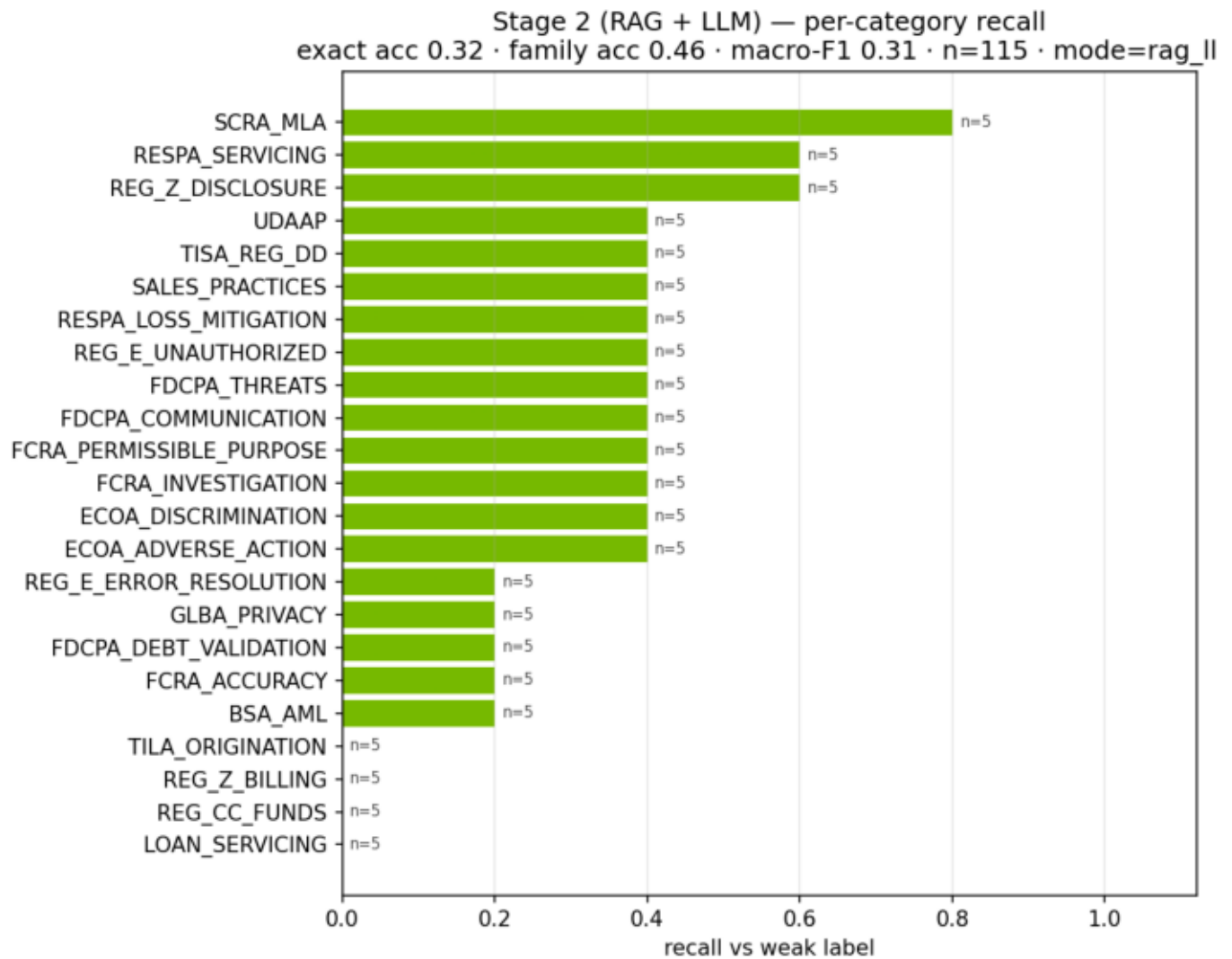
4 • Stage-1 ROC / PR curves



4 • Stage-1 confusion matrix



5 · Stage-2 per-category recall



Exact acc 0.3217 · family acc 0.4609 · macro-F1 0.3098 · n=115.

5 • Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.2
ECOA_ADVERSE_ACTION	5	0.4
ECOA_DISCRIMINATION	5	0.4
FCRA_ACCURACY	5	0.2
FCRA_INVESTIGATION	5	0.4
FCRA_PERMISSIBLE_PURPOSE	5	0.4
FDCPA_COMMUNICATION	5	0.4
FDCPA_DEBT_VALIDATION	5	0.2
FDCPA_THREATS	5	0.4
GLBA_PRIVACY	5	0.2
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_ERROR_RESOLUTION	5	0.2
REG_E_UNAUTHORIZED	5	0.4
REG_Z_BILLING	5	0.0
REG_Z_DISCLOSURE	5	0.6
RESPA_LOSS_MITIGATION	5	0.4
RESPA_SERVICING	5	0.6
SALES_PRACTICES	5	0.4
SCRA_MLA	5	0.8
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.4
UDAAP	5	0.4

Recall vs weak labels on the stratified evaluation sample.